BASEREAL-OS · ESSAY · VERSION 3

Inverse 3/2 Law

Geometric Interpretation of Kepler's Third Law

Version 3: the invariant of the drawing is the inverse-square law, and mass is the source of the field

$\alpha_n = 2\pi / n^{3/2}$ $L^2 r = 4\pi^2 R^3 = \mu T^2$ $a = \mu / r^2$ $M = \mu = \Phi / 4\pi$

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Project	BaseReal-OS https://vitalydecoder.github.io/BaseReal-OS/
Previous version	V2, 26 April 2026, https://doi.org/10.5281/zenodo.19929621
All versions	https://doi.org/10.5281/zenodo.19799606
Status	A geometric framework for building virtual universes; classical mechanics in geometric notation, not a claim of new physics

Abstract

The inverse 3/2 law is a single drawing: arcs of radius nR around a body, each opening by the angle $2\pi/n^{3/2}$, the angle that a body at distance nR sweeps during one surface period T of the central body (Kepler's third law). The drawing has an invariant: the square of any arc times its radius equals $4\pi^2R^3$; Version 2 called it the geometric mass of the body. Once the period T is added, each arc becomes the path travelled in one period, and the invariant turns into the inverse-square law: the pull toward the centre at distance r is $a = \mu/r^2$, with $\mu = 4\pi^2R^3/T^2$. Treating each arc as a meridian of a sphere, the flux of this pull through every sphere is the same, $4\pi\mu$. The source of the field is therefore μ itself, and the mass of a body is $M = \mu = 4\pi^2R^3/T^2$, set by two numbers: its radius R and its surface period T. The same period fixes the mean density, $\rho = 3\pi/T^2$, and mutual attraction obeys Newton's third law exactly. The geometric mass, the decoupling of mass and field, and the Earth "anomaly" of Version 2 are withdrawn; §8 states what changed and why.

1. Purpose

- A framework for virtual universes and educational simulators (in the spirit of Kerbal Space Program), in which each body is set by two physically meaningful numbers: its radius R and its period T.
- The framework is classical mechanics in geometric form. The derivations below reproduce Kepler's third law, Newton's inverse-square law and Gauss's flux theorem; nothing here claims new physics.
- In the units used here the gravitational constant is G = 1, so mass has dimension L^3T^{-2} (LT-System v2).

2. Construction

A vertical axis rises from a central point. Arcs of radius R, 2R, 3R, … start on the axis. The arc of radius nR opens by the angle $\alpha_n = 2\pi / n^{3/2}$. The first arc (n = 1) closes into a full circle; the arcs can be continued without limit (Figure 1).

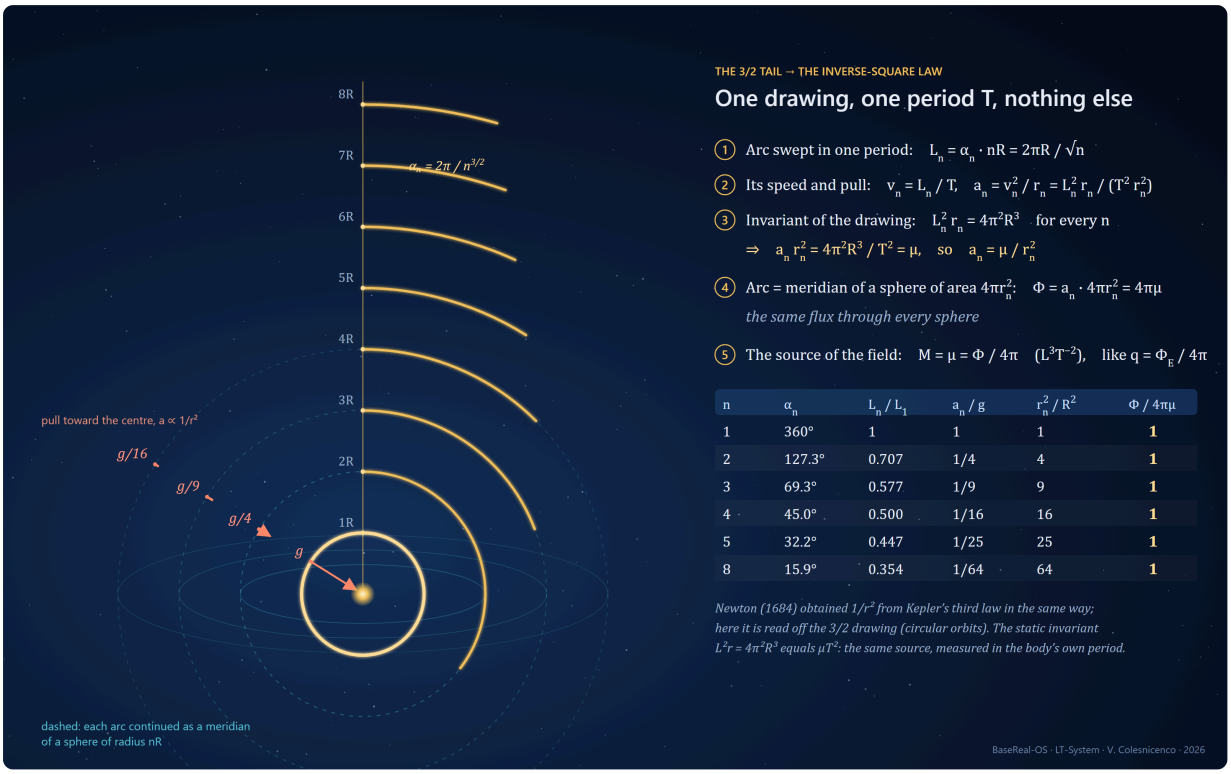


Figure 1. The 3/2 drawing (gold), its arcs continued as meridians of spheres (dashed), the pull toward the centre (g, g/4, g/9, g/16) and the derivation of §3–5. The last column of the table is the flux through each sphere: the same for every n.

3. The invariant of the drawing

The length of the arc of radius nR is $L_n = \alpha_n \cdot nR = 2\pi R / \sqrt{n}$. Its square times its radius is the same for every arc:

$L_n^2 \cdot r_n = (2\pi R / \sqrt{n})^2 \cdot nR = 4\pi^2 R^3$.

Version 2 interpreted this constant as an inertial parameter, the "geometric mass" of the body. Sections 4 and 5 show that it is the inverse-square law written in geometric form.

4. Adding the period: the inverse-square law

Let T be the period of an orbit at the surface of the body. Each arc is then the path travelled in one period by a body circling at distance nR: by Kepler's third law its period is $P_n = T \cdot n^{3/2}$, so in time T it sweeps the angle $2\pi / n^{3/2}$.

- Angular velocity $\omega_n = \alpha_n / T$, orbital velocity $v_n = L_n / T$.
- The field parameter is the same at every distance: $\mu = \omega_n^2 r_n^3 = v_n^2 r_n = 4\pi^2 R^3 / T^2$.
- The pull toward the centre (centripetal acceleration) at distance r_n: $a_n = v_n^2 / r_n = L_n^2 r_n / (T^2 r_n^2) = 4\pi^2 R^3 / (T^2 r_n^2) = \mu / r_n^2$.

The invariant of the drawing is therefore $L^2 r = a \cdot r^2 \cdot T^2$. Dividing it by T^2 gives the inverse-square law, $a \cdot r^2 = \mu$. At the surface (n = 1) the pull is $g = 4\pi^2 R / T^2$; at distance nR it is g / n^2.

This is how Newton obtained the inverse-square law from Kepler's third law for circular orbits (*De motu*, 1684); here it is read off a single drawing.

5. Arcs as meridians: the flux and the source of the field

Let each arc be a meridian of a sphere of radius r_n. The sphere has area $4\pi r_n^2$, and the flux of the pull through it is $\Phi = a_n \cdot 4\pi r_n^2 = 4\pi\mu$, the same for every sphere (Gauss's theorem).

The quantity that the field carries unchanged to any distance is μ. It is the source of the field, the mass of the body: $M = \mu = \Phi / 4\pi = 4\pi^2 R^3 / T^2$, with dimension L^3T^{-2}.

- With G = 1, Newton's law of gravitation has no constant: $F = M_1M_2 / r^2$, the same form as Coulomb's law $F = q_1q_2 / r^2$, in which a charge is likewise the flux of its field divided by 4π.
- The invariant of Version 2 survives under a new name: $4\pi^2R^3 = \mu T^2$, the same source measured in the body's own period.
- Kilograms are a conversion: $M[\text{kg}] = \mu / G_{SI}$, with $G_{SI} = 6.674 \cdot 10^{-11}$ m^3 kg^{-1} s^{-2}. Astronomy measures μ directly, and far more precisely than G or M separately.

6. Each body has its own time

The surface period follows from the measured radius and field parameter, $T = 2\pi\sqrt{(R^3/\mu)}$. The mean density follows from T alone: $\rho = M / V = (4\pi^2R^3/T^2) / (4/3 \cdot \pi R^3) = 3\pi / T^2$.

Body	R, km	T, s	$\rho = 3\pi/T^2 \div G_{SI}$, kg/m^3
Earth	6 371	5 061	5513
Moon	1 737	6 498	3344
Venus	6 052	5 190	5243
Mars	3 390	5 991	3934
Jupiter	69 911	10 319	1326
Sun	695 700	10 008	1410

The agreement with measured densities holds by construction (T is computed from the same R and μ): one number, T, carries the density of a body. Earth, Mars and the Sun come close to 5, 6 and 10 kiloseconds, but there is no pattern of round periods (LT-System v2, §7): any T is admissible.

7. Mutual attraction and Newton's third law

Each body pulls the other with its own field: $a_1 = M_2 / d^2$, $a_2 = M_1 / d^2$. The forces are equal and opposite, $F = M_1M_2 / d^2$, momentum is conserved, and the approach acceleration is $(M_1 + M_2) / d^2$. In an engine this is one rule per pair; no "dominant body" approximation is needed.

Example: Earth and Venus at $d = 4.2 \cdot 10^{10}$ m, the configuration used in Version 2.

Quantity	Value
Venus toward Earth, μ_{Earth} / d^2	$2.260 \cdot 10^{-7}$ m/s^2
Earth toward Venus, μ_{Venus} / d^2	$1.842 \cdot 10^{-7}$ m/s^2
Approach acceleration	$4.101 \cdot 10^{-7}$ m/s^2

Quantity	Value
Share of the approach, Venus / Earth	55.1 % / 44.9 %
Force on each planet, $\mu_{Earth}\,\mu_{Venus} / d^2$	$7.34 \cdot 10^7$ m^4/s^4 ($1.10 \cdot 10^{18}$ N after dividing by G_{SI})

8. What changed relative to Version 2

Item	Version 2	Version 3
The invariant $L^2r = 4\pi^2R^3$	the geometric mass of the body, an inertial parameter, pictured as equal-volume torus sectors (585 kg per m^3)	the inverse-square law in geometric form; equals μT^2; the torus picture is removed
Mass	$M = 4\pi^2R^3$ (L^3)	$M = \mu = 4\pi^2R^3/T^2$ (L^3T^{-2}), the flux of the field divided by 4π
Field and mass	decoupled; "a gain of several percent for small bodies"	one quantity; the claim is withdrawn
Earth period	"anomaly +60 s": a radius 50 km larger after the Theia impact	withdrawn: T = 5061 s follows from the measured R and μ and reflects the mean density
Mutual attraction	one acceleration split by geometric masses (54 % / 46 %)	each body pulls with its own μ; exact (55 % / 45 %)
Inverse-square law	"naturally follows"	derived step by step from the drawing (§4) and closed by the flux through spheres (§5)

Why the geometric mass had to go. With $M = 4\pi^2R^3$ every body has the same density, $M/V = 3\pi$: calibrated on the Earth, every body receives the Earth's mean density of 5513 kg/m^3. Mars then comes out 1.4 times too heavy, the Sun 3.9 times, and the Earth/Moon mass ratio is 49.3 instead of the measured 81.3. Dividing the same $4\pi^2R^3$ by T^2 gives each body its own density and the measured masses.

9. For virtual universes

The aim is unchanged: a world is built from two numbers per body, R and T. The field, the orbits ($P = T \cdot (a/R)^{3/2}$), the surface gravity ($g = 4\pi^2R/T^2$), the mean density ($\rho = 3\pi/T^2$) and the mutual attraction all follow from them, with no gravitational constant and no density model. Users can create stars, planets and moons with physically accurate mechanics; the interactive demonstrations of the BaseReal-OS project use exactly these rules.

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